Department Overview Department Name: Department of Civil Engineering Faculty: Faculty of Engineering & Architecture Accreditation & Framework: Fully accredited by the Pakistan Engineering Council (PEC). The department implements Outcome Based Education (OBE) and Outcome-Based Assessment (OBA) models at the undergraduate level as recommended by the Engineering Accreditation Board (EAB) of the PEC. History: Actively contributing to educational infrastructure and academia since 2008. The postgraduate (MS) program was introduced in 2015. Research Focus Note: Highlights an active engagement in Affordable Housing Research aligning with national construction organizations aimed at providing housing facilities to homeless populations and boosting economic activity in Pakistan.

Mission To produce competent professionals of high ethical values, enabling them for executive positions in civil engineering, higher studies in prestigious universities, life-long learning, and societal leadership. Also, to provide a dynamic learning environment that emphasizes sustainable engineering practices, problem-solving skills, teamwork, communication, and promotes cutting-edge research and leadership skills.

Program Educational Objectives (PEOs) Civil engineering graduates are expected to demonstrate the following attributes 3 to 5 years post-graduation:

1. PEO-1: Essential knowledge and adequate civil engineering skills to meet industrial requirements.
2. PEO-2: High ethical values and societal responsibilities for sustainable development.
3. PEO-3: Effective communication, strong interpersonal skills, and the capacity to lead and work collaboratively in teams.
4. PEO-4: Ability to keep abreast with the latest innovations and technological developments through continued professional practices, higher studies, and research.
5. Bachelor of Science in Civil Engineering (BS Civil)

Admission Requirements • Academic Background: F.Sc (Pre-Engineering) from a recognized board or an equivalent certification with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in the relevant field with a minimum of 60% marks. • Testing: GAT General or the university's internal NTS test.

Degree Requirements • Total Credit Hours: 136 • Total Courses: 46 • Minimum Academic Standing: CGPA $\ge$ 2.0

Curriculum Plan (Program Schema by Semester) Note: Credit hour listings follow the format (Theory + Lab = Total).

1st Semester • CIVILE-206 | Engineering Materials | Prerequisite: None | Credit Hours: 2+1=3 • ENGG-201 | Engineering Drawing | Prerequisite: None | Credit Hours: 1+2=3 • HUM-161 | Reading & Writing Skills | Prerequisite: None | Credit Hours: 2+0=2 • MATHA-117 | Applied Calculus | Prerequisite: None | Credit Hours: 3+0=3 • HUM102-/101 | Pakistan Studies / Islamic Studies or Ethics | Prerequisite: None / None | Credit Hours: 1+0=1 / 2+0=2

2nd Semester • CIVILE-207 | Engineering Surveying-I | Prerequisite: None | Credit Hours: 2+1=3 • GEOE-302 | Engineering Geology | Prerequisite: None | Credit Hours: 2+1=3 • ENGG-205 | Engineering Mechanics | Prerequisite: None | Credit Hours: 3+1=4 • MATHA-214 | Differential Equations | Prerequisite: Applied Calculus | Credit Hours: 3+0=3 • ENGG-204 | Basic Electro-Mechanical Engineering | Prerequisite: None | Credit Hours: 2+1=3 • CIVILE-303 | Civil Engineering Drawing & Graphics | Prerequisite: Engineering Drawing | Credit Hours: 1+2=3

3rd Semester • CS-114 | Programming Fundamentals | Prerequisite: None | Credit Hours: 2+1=3 • CIVILE-314 | Engineering Surveying-II | Prerequisite: Engineering Surveying-I | Credit Hours: 2+1=3 • CIVILE-302 | Mechanics of Solids-I | Prerequisite: Engineering Mechanics | Credit Hours: 2+1=3 • Econ-105 / ENGG-206 | Engineering Economics / Fluid Mechanics-I | Prerequisite: None / None | Credit Hours: 2+0=2 / 3+1=4 • CIVILE-304 | Quantity & Cost Estimation | Prerequisite: None | Credit Hours: 2+1=3

4th Semester • CIVILE-419 / 208 | Construction Engineering / Structural Analysis-I | Prerequisite: None / Engineering Mechanics | Credit Hours: 3+0=3 / 3+0=3 • CIVILE-306 | Soil Mechanics | Prerequisite: None | Credit Hours: 3+1=4 • MATHA-216 | Numerical Analysis | Prerequisite: None | Credit Hours: 3+0=3 • ENGG-321 | Environmental Engineering-I | Prerequisite: None | Credit Hours: 2+1=3 • CIVILE-208 | Concrete Technology | Prerequisite: None | Credit Hours: 1+1=2

5th Semester • MATHA-217 | Probability and Statistics | Prerequisite: None | Credit Hours: 2+1=3 • ENGG-209 | Fluid Mechanics-II | Prerequisite: Fluid Mechanics-I | Credit Hours: 3+1=4 • HUM-268 / MGMT-304 | Communication & Presentation Skills / Org. Behaviour | Prerequisite: None / None | Credit Hours: 2+0=2 / 2+0=2 • CIVILE-309 / 310 | Reinforced Concrete Design-I / Transportation Planning | Prerequisite: None / None | Credit Hours: 3+0=3 / 3+0=3

6th Semester • ENGG-441 | Construction Management | Prerequisite: None | Credit Hours: 2+1=3 • CIVILE-418 | Mechanics of Solids-II | Prerequisite: Mechanics of Solids-I | Credit Hours: 2+1=3 • CIVILE-403 | Highway and Traffic Engineering | Prerequisite: Transportation Planning | Credit Hours: 3+1=4 • CIVILE-402 / 315 | Engineering Hydrology / Structural Analysis-II | Prerequisite: None / Structural Analysis-I | Credit Hours: 2+1=3 / 3+0=3

7th Semester • HUM-276 | Technical Report Writing | Prerequisite: None | Credit Hours: 0+1=1 • MGMT-315 / 311 | Entrepreneurship / Architecture & Town Planning | Prerequisite: None / None | Credit Hours: 3+0=3 / 3+0=3 • CIVILE-404 | Geotechnical and Foundation Engineering | Prerequisite: Soil Mechanics | Credit Hours: 3+1=4 • CIVILE-407 | Reinforced Concrete Design-II | Prerequisite: Reinforced Concrete Design-I | Credit Hours: 3+1=4 • CIVILE-406L | Civil Engineering Project-I | Prerequisite: None | Credit Hours: 0+3=3

8th Semester • CIVILE-401 | Geoinformatics | Prerequisite: Engineering Surveying-II | Credit Hours: 1+1=2 • CIVILE-412 / 411 | Environmental Engineering-II / Steel Structures | Prerequisite: Environmental Eng.-I / None | Credit Hours: 2+0=2 / 3+0=3 • CIVILE-410 | Hydraulics and Irrigation Engineering | Prerequisite: Fluid Mechanics-II | Credit Hours: 3+1=4 • HUM-112 | Professional Ethics | Prerequisite: None | Credit Hours: 2+0=2 • CIVILE-417L | Civil Engineering Project-II | Prerequisite: None | Credit Hours: 0+3=3

2. Master of Science in Civil Engineering (MS Civil)

Admission Requirements • Academic Background: 16 years of education or an equivalent benchmark qualification (e.g., a 4-year B.E. or BS degree) in Civil Engineering from an HEC-recognized institution. • Minimum Marks: At least 60% marks under the annual grading setup, or a minimum program CGPA of 2.5 out of 4.0. • Testing: A valid GAT General result with a passing score of at least 50% marks.

Degree Requirements • Total Credit Hours: 30 • Total Coursework Requirements: 8 advanced postgraduate courses. • Minimum Academic Standing: CGPA $\ge$ 2.5